

Informal and Formal Channels in Boundary-Spanning Communication

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Boundary-spanning communication has been found to be important to various types of organizations and groups as a source of new information and awareness of environmental changes. Although the importance of formal, public channels in building the knowledge base of a field has long been recognized, most studies of boundary-spanning activity have been limited to direct, personal (informal) communication. This study examines both informal and formal communication patterns of three groups in order to determine whether, in fact, a variety of media are used in boundary-spanning communication. The groups selected were the members of three professions which fill intermediary positions between the creators and consumers of cultural products; one located in academia and two in the private sector. Major findings were that: (1) A boundary-spanning structure linking 82.2% of the survey respondents was composed of both informal and formal media. (2) The professions differed in which type of channel was most heavily used, but were similar in their perceptions of the importance of the channels. (3) Individuals central to the informal part of the structure were also more likely to use the formal parts of the structure. (4) Boundary-spanning media were among the five most frequently cited media for all three of the professions.

Introduction

The intent of this study is to examine the phenomenon of boundary-spanning communication to determine the relative importance of the various channels through which work-related information can be brought into a group. Any group can be considered as an information system, with ideas and information moving within its boundaries, and also being imported from outside. Information from outside the system has particular potential for having an impact, as it can lead to alternations in the assumptions and knowledge which form the basis of the group's activities. A group's vitality may be increased by awareness of innovations, alternative approaches to problems, ideas and values not

previously considered, and external factors which may have impacts on the group.

Factors such as economic relationships, geographic proximity, and intellectual proximity influence which outside individuals and groups are chosen for communication. The groups selected for this study were three professions which have related intellectual interests but which fulfill different societal functions. Although not linked as employees of a company are by physical proximity or a hierarchy of control, members of a profession constitute an identifiable population of individuals united by job title, the nature of the work performed, educational preparation, and the exchange of information and ideas about their field. They work within defined intellectual and task-related milieus.

Most boundary-spanning research has focused on direct personal communication. However, it is the contention of this article that substantive intellectual boundary spanning also occurs when members of different groups share formal communication channels, such as journals. This study was designed to determine whether both informal and formal boundary-spanning activities did occur, and if so, whether both have importance as measured by the number of individuals involved and the individual's assessments of their impacts.

Background

Of interest first is existing research which examines the relationship of boundary-spanning communication to organizational vitality. Then, because the structure along which boundary-spanning communication occurs may include both informal (personal) and formal (public) components, research into informal and formal communication will be reviewed. Most, though not all, of this research has examined information transfer within group boundaries, particularly scientific disciplines.

Research in the fields of business and management demonstrates that communication across system boundaries can have significant impacts on the health of the system.

Two major effects of boundary-spanning communication have been identified. One is that such activity

Received March 7, 1990; revised June 14, 1990 and November 1, 1990; accepted July 9, 1991.

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contributes to variation within the system, thus permitting adaptive behavior within its ecological niche (Aldrich, 1979). Ritvo, Salipante, and Notz (1979) studied governing boards of hospitals and found that those which used data originating outside the hospital were faster than those relying solely on internal data to recognize the financial problems they would face as the result of nationally declining birth rates. Tushman (1979) cites evidence that scientists in R & D labs found scientists outside their own labs more useful than those inside for generating new ideas, though sources within their own labs were more useful for implementation and problem-solving information. More recently, Dollinger (1984) found a relationship in entrepreneurs between boundary-spanning activities and healthy financial performance of the business. Thus there is strong evidence that the variation introduced by the transfer of information across system boundaries can contribute to innovation, to successful response to changes in the environment, and to organizational vigor.

The reciprocal effect of such boundary spanning communication is that as the variation within systems increases, the variation between them may decrease. Galaskiewicz (1985) found in a study of corporate giving officers in 150 companies, that as they exchanged information with each other they came to share each others' assessments of the nonprofit organizations which solicited funds from them. The more proximate two officers were in the communication network, the more likely they were to evaluate prospective donees in the same way.

Much of the literature on informal communication has been concerned with networks linking individuals in various populations. Rogers and Kincaid (1981), among many others, assert that network structure is an essential variable in understanding group and system performance. Williams, Rice, and Rogers (1988) discuss the ways in which network analytic techniques reveal patterns of relationships which are interactive in nature. Wellman and Berkowitz (1988) argue the importance of the fact that "individuals, groups, and organizations ... are bound up with one other, simultaneously, in a number of quite different ways," and that the relationships between them shape the patterns of behavior in which they engage.

One type of network is the social circle. A social circle is defined as a set of individuals connected to one another by some form of social choice (based on Kadushin, 1970). Its existence is determined by asking the individuals under study for the names of the people with whom they have interactions of some specified nature. Different kinds of networks are identified depending upon the types of questions asked. The resulting group of individuals are linked to one another either directly or indirectly. A direct link exists when one individual names another; an indirect link exists when two individuals name or are named by a third individual, but do not name each other. Thus, an indirect link

exists between individuals A and C in any of the following situations: (1) A names B and B names C; (2) A names B and C names B; (3) B names A and C; and (4) C names B and B names A. The significance of the existence of a social circle is that individuals who discuss their work with one another can be assumed to communicate a portion of their knowledge, ideas, and attitudes to each other; in addition, the knowledge, ideas, and attitudes of the other people with whom they discuss their work may also reasonably be assumed to be communicated.

Warren Hagstrom's classic *The Scientific Community* (1965) argued that the central goals and values of a scientific community are expressed and established primarily through such informal communications. In addition, informal networks of individuals were shown to serve a vital function in stimulating new ideas and developing existing ideas. John Ziman, in *Public Knowledge* (1978), stressed the importance of informal communications for discussing scientific ideas while they are still incomplete and untidy, before they are ready to be formalized in a published article. William Garvey (1979) determined that similar informal networks exist in the social sciences, though they are less highly structured than those of physical scientists. Derek de Solla Price (1983) argued in *Big Science, Little Science* that social organization among scientists is highly developed because personal communications are more effective than published communications for establishing intellectual property rights over one's work.

Diana Crane (1972) looked at both communication within research communities and communication between them. She found that while researchers tended to have numerous contacts of both types, the within-group contacts tended to converge on a small group of individuals (an invisible college), while there was little overlap of the individuals from outside the group who were contacts of group members.

Formal communication has also been an important area of study. Formal communication is central to any well-developed field; it constitutes the official statement of the knowledge base of a field. In the sciences, the refereed journal article is the recognized medium for reporting results and discoveries. John Ziman's (1978) *Public Knowledge* asserts that the fundamental nature of science lies in the making public of new knowledge in order that it may be challenged and tested. In the social sciences and humanities, journal literature is important as well, and books have an importance that is greater than in the sciences (Hagstrom, 1965). In the professions, formal communications also provide statements of current belief, newly discovered information, personal experience, or suggestions for successful completion of tasks. The editorial process involved in formal communication ensures some level of quality assessment and control, though the range is extremely wide.

Meetings of professional and scholarly associations provide another formal channel of communication. Pa-

pers may be invited or submitted for refereeing, and those selected are then presented to those attending the conference. In some cases, the papers are then published in proceedings volumes. Although many informal interactions take place at conferences, this study considers them as formal communication channels since the basic nature of a conference is that it is formally convened and the content is planned.

As Rice and Shook (1990) point out, "most studies of organizational media use treat each communication channel in isolation, rather than identifying underlying patterns." The same has been true in many studies of scholarly communication. Those studies which examine both informal and formal communication indicate that both have importance for the transmission of ideas.

In a discussion of a controversy in oxidative phosphorylation research, Kemp (1977) analyzed the complementary use of informal and formal communication channels for debate. Formal channels gave the widest access for the scientists' arguments. Journal publication was used to establish position within the research specialty, and a lecture delivered to an association of the discipline provided an opportunity to persuade a larger scientific community. At the same time that the debate was being waged through formal channels, the two scientists carried on a vigorous informal exchange through letters, often over 1000 words in length, which were, in turn, circulated to 41 other interested scientists. In both types of communication, the scientists used tactics of criticism, justification, and persuasion to achieve a consensus of opinion.

During the years of the public debate over the U.S. involvement in Vietnam, Kadushin (1974) conducted a study of American intellectuals. He found that 16% identified journals as being the major influence on their thinking about the war in Vietnam, and 20% said that direct discussions with friends and acquaintances had had the most effect; another 13% cited the general sense of the intellectual climate of the time, presumably an extrapolation and generalization from both informal and formal communications.

Several studies have shown that heavy users of one channel type are also heavy users of the other. Allen (1970), for instance, found that the scientists and engineers who were the most important in transferring information across organizational boundaries were also more widely read than their colleagues.

Zaltman (1974) studied theoretical high-energy physicists; he reported that the most highly connected individuals placed a higher value on a greater number of information channels than did those who were less central. Of the 11 channels about which they were asked, the three rated most highly were conversations with others at their own institutions, conversations with people at other institutions, and journals.

The research reviewed here highlights the importance of boundary-spanning communication in introducing new information into a system, and the importance

of both informal and formal media as channels through which ideas are transmitted. The current study will examine three professional groups which have some intellectual concerns in common to determine whether boundary-spanning communication occurs between them and, if so, through what channels.

The Study

Three professions were selected which were thought to be closely enough related that it would be possible for them to share some formal and informal communication channels. The three professions were those of editor, book reviewer, and scholar of children's literature.

Precisely speaking, though the occupations studied here would be referred to in everyday speech as professions, only one of the three meets the criteria used by sociologists of professions to define a true profession. The other two would be referred to as semiprofessions (Etzioni, 1969), meaning not that they are subsets of some other occupation, but that they have not fully developed all of the attributes which constitute the definition of professional status. These attributes include an abstract and organized body of knowledge, recognition by society that members of this occupational group possess this knowledge and that certain problems should therefore be given over to them for solution (for example, health problems to doctors), and an ideal of service. For the sake of simplicity, the profession and semiprofessions discussed here are referred to as professions or professional groups, without distinction.

These three professions function as cultural intermediaries; the nature of their work places them between the creation of works of art and the reception of those works by an audience. Other examples of such cultural intermediaries are art gallery owners and editors of journals, who select from original works those which will be displayed or published. Editors, reviewers, and scholars are particularly interesting as intermediaries because their jobs explicitly involve evaluating literature and applying aesthetic standards, but in different ways and for different purposes. They work with the same cultural artifacts, books for children. Yet the nature of the work of each profession is quite different from the others.

Editors make the fundamental decision about which artistic creations will actually reach the public in the form of published books. They must necessarily think of children's books both as works of art and as products which will be sold. Their evaluations and analyses are directed primarily to the author and illustrator in the editorial process, and also to other publishing professionals as decisions are made about which manuscripts to publish, how best to promote them, etc.

Reviewers see a book or galley proofs after the decision to publish has been made; their evaluations are directed specifically to individuals making book-purchase decisions. Their evaluations are written and published

in review journals which, because of the nature of the market for children's books, are generally read by other cultural intermediaries—librarians and booksellers—not by the ultimate consumer.

Scholars write about whatever literature is of interest to them, whether new or centuries old. Their purpose is not to provide information for decision making, but to reflect upon and analyze some aspect of the work of literature which will enhance understanding of that work. The majority of scholars of works for children are professors in university English literature departments. Their analyses are presented in scholarly journals and books, in papers at conferences, and in the classes they teach.

These three professions, then, represent two different sectors of the economy, the corporate world of business and academia. They share subject matter and an analytic/evaluative approach to it, but the products of their work are different and their own audiences are different. It would be expected that the three groups of individuals might share some communication channels and have others that were specific to each profession.

Methodology

The three populations were carefully and narrowly defined.

Editors included were those doing substantive editing (not copyediting) for publishing companies with clearly identified juvenile lists.

Reviewers were defined as individuals who were employed by review journals and/or listed on the masthead of the journal. Individuals who wrote occasional reviews and freelance reviewers were not included.

The scholars presented the largest problem of definition. They cannot be always be identified either through their job titles or their places of employment. Further, not all thoughtful considerations of children's literature in essay or book form can be considered "scholarship." Therefore, scholars working in the field of children's literature were identified through a two-step process. A sample of members of the Children's Literature Association (the goal of which is to encourage serious scholarship and research in children's literature) was asked to identify those journals which they considered to be important publishers of criticism concerning children's literature. Individuals who had published an article or had a book of criticism reviewed between 1980 and 1985 in those journals then formed the population of scholars.

Questionnaires were mailed during the 1985–1986 academic year to all identified members of the populations. One follow-up letter and questionnaire were sent. Both the sizes of the populations and their response rates varied. Of the 168 editors, 26.2% (44) returned the survey forms. Although this was the lowest of the three response rates, that limitation is mitigated somewhat by the fact that the respondents were employed in 23 of the 44 publishing companies, so 52.3% of the

companies are represented in the results. In addition, respondent and nonrespondent editors were very similar in terms of job status and size of company. The critics' response rate was 47.3%, 123 of the 260 identified. The response rate for the reviewers was 68.4%; 19 were identified and 13 responded.

Questions in the survey instrument were designed to identify informal and formal communication channels used by respondents. Respondents were asked to list journals read most often, associations and organizations in which they held memberships, and conferences and book fairs which they found most interesting or useful to attend. These questions permitted identification of the formal channels used and the numbers of respondents using them. To determine informal channels and the number of respondents linked by them, respondents were asked to list the individuals whom they knew personally, with whom they discussed their work or from whom they obtained information for their work. The question was then rephrased to specify individuals belonging to the two professional groups to which the respondent did not belong. The responses to these questions allowed detection of interpersonal networks among the respondents.

The focus of this article is the boundary-spanning communication which links these three professions. Although each had an interesting internal communication structure and numerous links to other professions and disciplines as well, these are not considered here. [For a discussion of the journals of several disciplines which were important to the scholars, see Weedman (1986).] The intent is to discover whether three groups of cultural intermediaries which deal with the same cultural artifacts but in extremely different ways and for different reasons, share a communication structure which permits the transfer of cultural information and ideas across the professional boundaries which separate them. Further, because both informal and formal channels are known to be important in many communication settings, the study was intended to discover whether both types of channels are important for this boundary-spanning communication.

The importance of the informal and formal channels was assessed through two partial measures. First, the questions described above permit a tally of the total number of respondents using and, therefore, linked by each channel. Second, a bank of questions in the instrument asked respondents to rate the importance of personal discussion, reading of professional and scholarly journals, and attendance of conferences and book fairs. Importance was defined in terms of stimulation and of the respondent's thinking and work with children's books: a scale from 5 (highly important) to 1 (not at all important) was used.

The data gathered also permitted exploration of three additional dimensions of the boundary-spanning structure: the extent to which the structure functioned differently for the three professions, the differences in

usage of the structure by central and peripheral individuals, and the use of the boundary-spanning media compared to the use of group specific media.

Findings

The Existence and Composition of a Boundary-Spanning Communication Structure

Communication across professional boundaries did indeed occur, and via both informal and formal channels. An informal social circle was found to exist and crossed the professional boundaries. Two types of formal channels provided boundary-spanning communication, journal literature, and conferences or book fairs sponsored by professional and scholarly associations. Twelve journals were read on a regular basis by at least one member of each of the three professional groups. Seven conferences and book fairs were listed by at least one member of each of the groups as being interesting or useful to attend.

Although a citation by one member of each group does literally constitute boundary spanning communication, it must be argued that such a low incidence of use indicates a very minor role in the structure of communication linking the groups. To filter out noise from such low-use channels, a threshold of 25.0% of all respondents was set as the point at which a given commu-

nication channel would be considered to provide a significant link among the groups. Those channels which linked 25.0% or more of all respondents, then, were considered to comprise the communication structure.

Four channels met this criterion. One was informal in nature, the social circle; it united 74.4% of the respondents. Three were formal, two journals and the conferences of a professional association. *The Horn Book Magazine* was read regularly by 44.0% of the respondents, *School Library Journal* was read regularly by 25.5% of the respondents, and the conferences of the American Library Association were considered interesting or useful to attend by 27.8% of the respondents.

Those respondents utilizing any aspect of the boundary-spanning communication structure will be referred to as "cosmopolitans" to distinguish them from respondents using only group-specific media. This term was derived from Pelz and Andrews's use of the terms "cosmopolitan" and "local" to differentiate between individuals oriented to their professions and those oriented primarily to the organization employing them (Pelz & Andrews, 1966).

The communication structure united 148 cosmopolitans, 82.2% of the 180 respondents; 17.8% were locals entirely outside the structure. Figure 1 shows the number of respondents using each element of the structure; it provides a starting point for understanding the importance of the different channels.

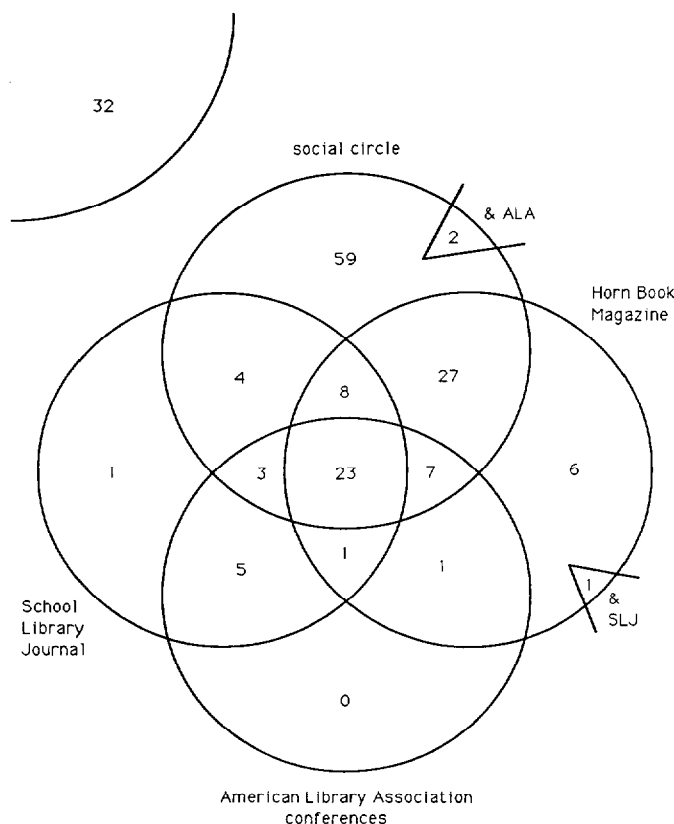


FIG. 1. The communication structure: Number of respondents in each sector.

The social circle is clearly the largest component, uniting 88.5% of the cosmopolitans. Next is *The Horn Book Magazine* (50.0%), *School Library Journal* (30.4%), and conferences of the American Library Association (27.0%).

Importance of the Informal and Formal Components of the Communication Structure

As discussed at the beginning of this article, no absolute measure of importance is possible. Partial measures are provided, however, by (1) analysis of the numbers of individuals linked by the structure and (2) their perceptions of the various communication channels' value. Subsequent sections will consider three additional dimensions of the channels' importance.

Numbers of Individuals Linked by Formal and Informal Channels. Precisely one-half of the 148 cosmopolitans used both formal and informal components of the communication structure; half used one or the other exclusively. Figure 2 illustrates this data. Informal channels were, by far, the most likely to be the exclusive boundary-spanning medium; 39.9% of the cosmopolitans cited none of the formal channels but were members of the social circle. Formal channels were more rarely used exclusively; only 10.1% of the cosmopolitans cited a formal channel but were not members of the informal social circle.

None of the formal channels considered individually functioned as the sole boundary-spanning medium for very many people; *Horn Book*, the most frequently cited, was the sole boundary-spanning medium for only 4.0% of the cosmopolitans. *School Library Journal* was the only boundary-spanning channel for one individual,

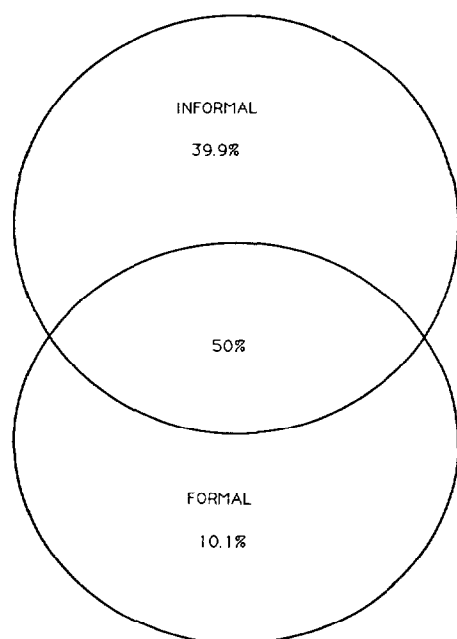


FIG. 2. Informal and formal components of the communication structure: Percentage of cosmopolitans using each.

and none of the cosmopolitans attended ALA conferences without using any of the other channels.

Perceived Importance of Informal and Formal Channels. Respondents were asked to rate the importance for their thought and work of various communication channels according to the following scale: 5 = highly important, 4 = important, 3 = somewhat important, 2 = minimally important, 1 = not at all important. The cosmopolitan's mean ratings of the types of communications channels which formed the boundary-spanning structure were all above 3; the mean for conferences and book fairs was 3.36, for professional or scholarly journals was 3.80, and for personal discussions was 4.09. The perceived importance of informal communication is further highlighted by the fact that only 5.4% of the cosmopolitans rated it in one of the bottom two categories; by contrast, nearly twice as many cosmopolitans (10.2%) rated journal reading in the last two categories, and 24.4% rated conference attendance there. Table 1 shows the perceived importance of the three boundary-spanning channels.

Differences Between the Professions

Reviewers and editors relied on both informal and formal components; 100% of the reviewers used both, and 67.6% of the editors. Critics were more likely, however, to use only the informal components—only 35.5% used both, while 57.1% belonged to the social circle and used none of the formal boundary spanning media. In contrast to the critics, of the editors who used only one or the other, the largest group was composed of those who used only the formal channels, 21.6%. Table 2 displays these differences, which are statistically significant.

TABLE 1. Perceived importance of the three boundary-spanning channels.

Channel	Rating					Mean
	1	2	3	4	5	
Discussion	.7%	4.7%	18.2%	37.2%	39.2%	4.09
Journals	3.4%	6.8%	26.0%	33.6%	30.1%	3.80
Conferences	9.5%	14.9%	27.0%	27.7%	20.9%	3.36

TABLE 2. Use of informal and formal channels by the three professions.

	Editors	Reviewers	Critics
Formal and informal	67.6% (25)	100% (13)	35.7% (35)
Formal only	21.6% (8)	.0% (0)	7.1% (7)
Informal only	8.1% (3)	.0% (0)	57.1% (56)
All four channels	29.7% (11)	84.6% (11)	1.6% (2)
	<i>n</i> = 37	<i>n</i> = 13	<i>n</i> = 98

$$X^2 = 42.42, 4 \text{ df}, p = 0.0000.$$

Figures 3(a–c) show the percentage of cosmopolitans in each professional group using each component of the communication structure. *The Horn Book* and *School Library Journal* were of almost equal importance for both editors and reviewers, though for editors who used only one of the four channels, *The Horn Book* was more frequently cited than *School Library Journal*. For critics, the two journals were not of comparable importance; *The Horn Book* was used by 39.8% of the critics, whereas *School Library Journal* was used by only 11.2%. Reviewers made more use of ALA conferences than the other two groups; 92.3% of the reviewers cited this element of the structure. Though less used by editors, the conferences were still cited by 51.3%. ALA conferences were cited by 10.2% of the critics.

In their ratings of the importance of the types of communication, the three groups were similar. Although reviewers and editors rated informal personal discussion more highly than critics, all three rated this category higher than either of the formal channels. The differences in the groups' ratings of discussions were statistically significant, but the differences between their ratings of the formal channels were not. Table 3

shows the mean rating assigned by each professional group to the channels.

Differences Between Central and Peripheral Individuals

Research in scientific communication indicates that individuals who occupy key positions in an informal communication network also make heavier use of formal media (Allen, 1969).

Two groups of respondents may be considered to occupy key positions in the social circle, those who themselves have direct links to individuals across the professional boundaries and those identified as members of the social circle's invisible college. These key or central individuals were compared to noncentral (here labeled "peripheral") social circle members in their use of formal and informal channels. Table 4 displays this information.

Both formal and informal boundary-spanning channels were used by almost three-fourths of the central individuals; only 28.3% used informal channels only (by definition, these individuals could not use formal

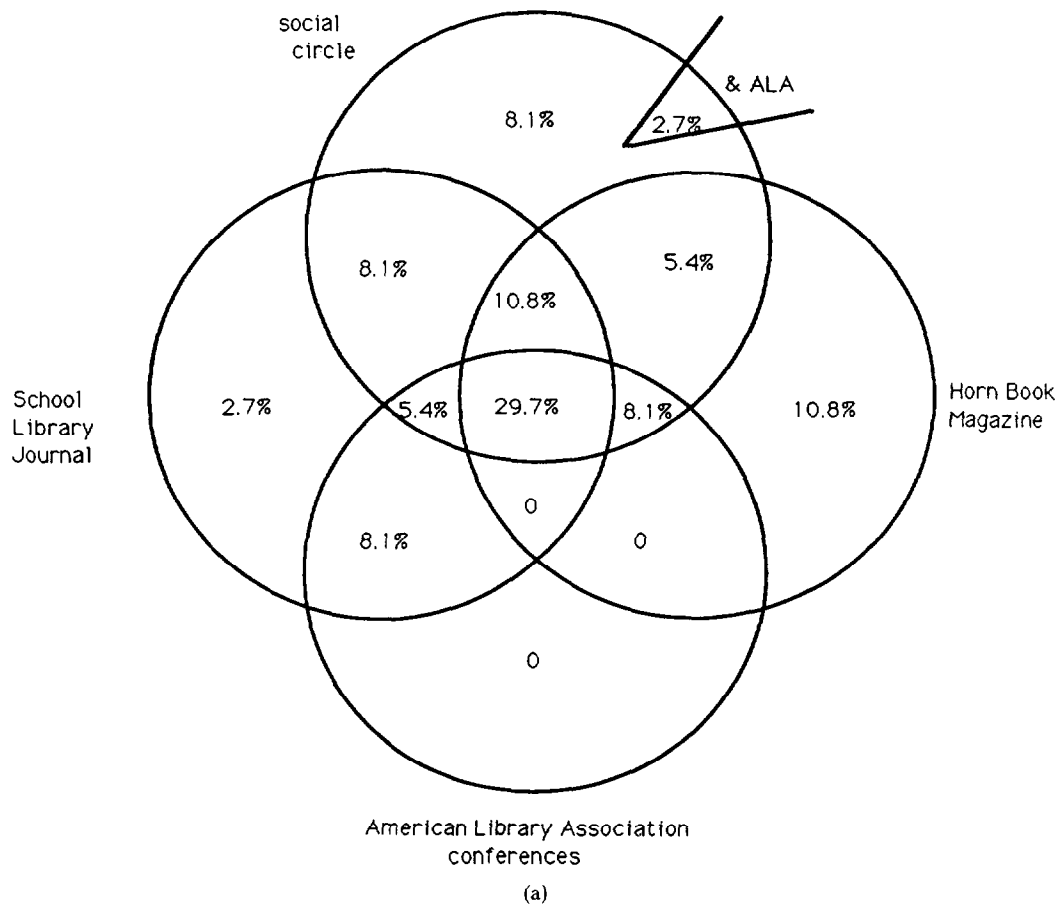


FIG. 3. (a) Percentage of editor cosmopolitans using the components of the communication structure ($n = 37$). (b) Percentage of reviewer cosmopolitans using the components of the communication structure ($n = 13$). (c) Percentage of critic cosmopolitans using the components of the communication structure ($n = 98$).

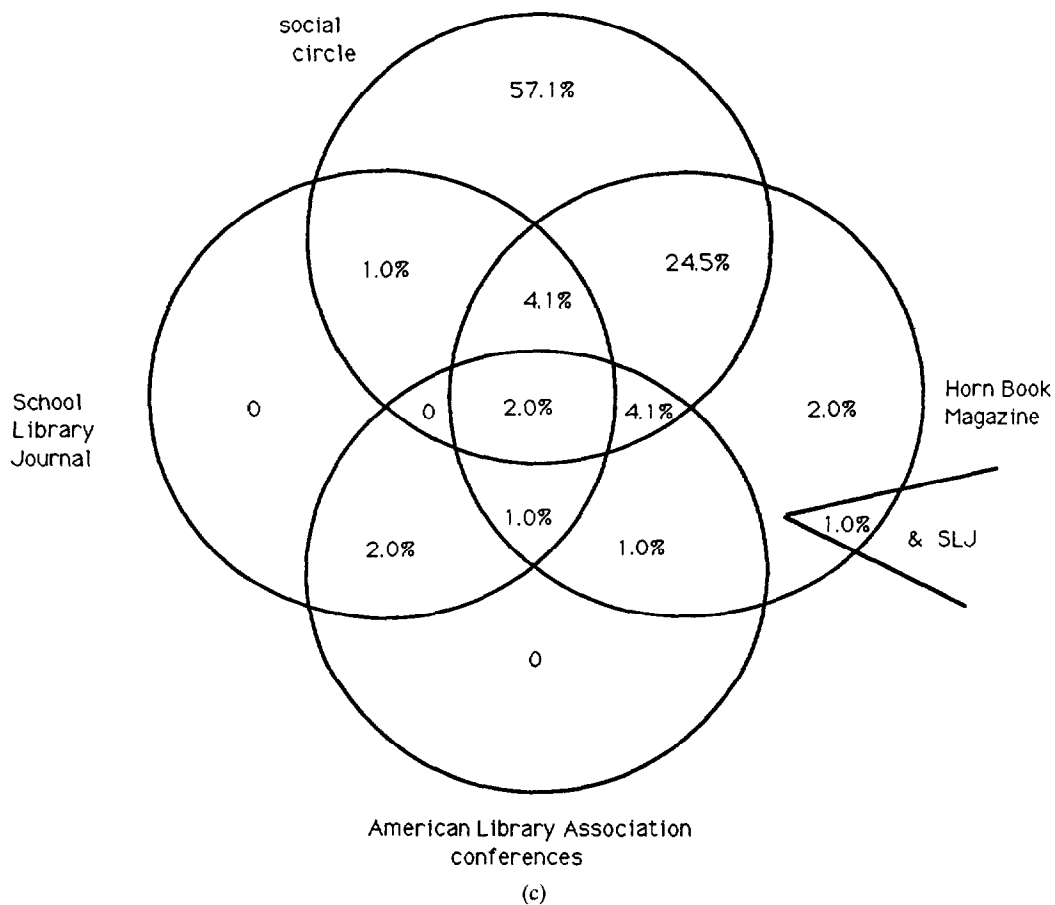
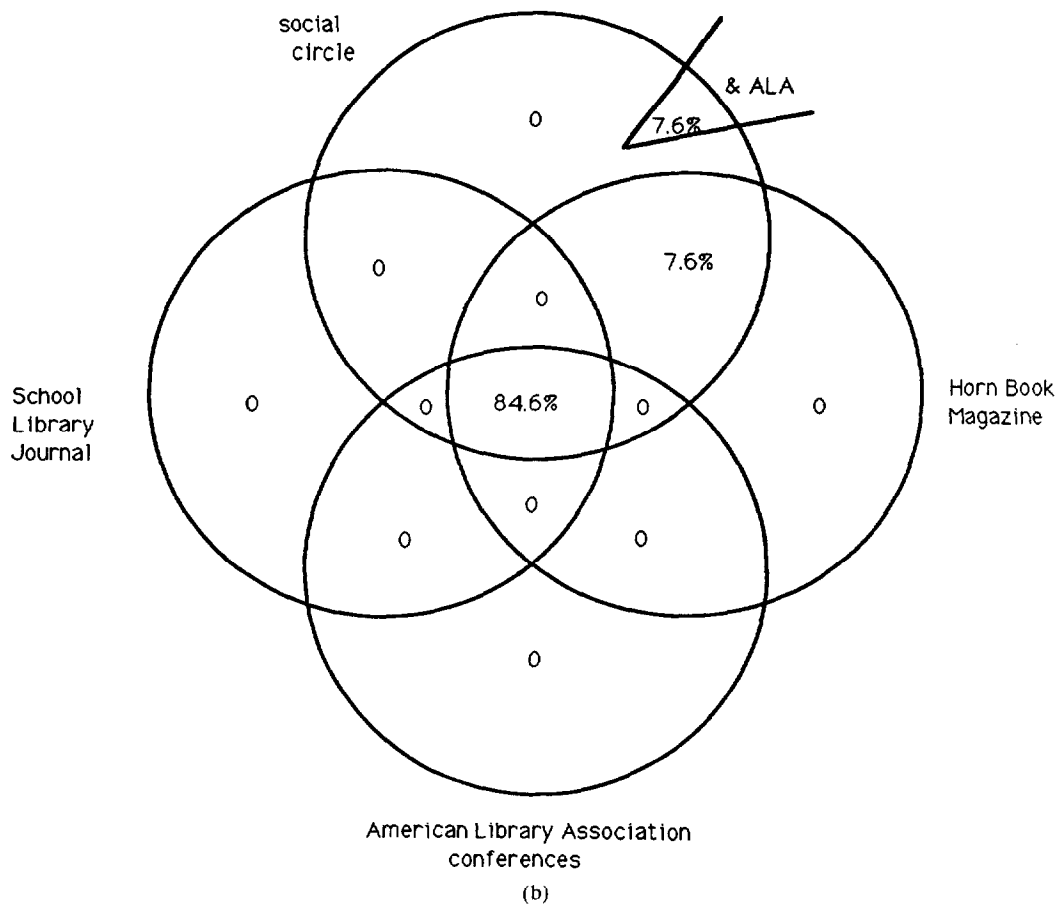


FIG. 3. (continued)

TABLE 3. Mean ratings of informal and formal channels, by professional group.

Channel Type	Professional Group			Significance
	Editors (<i>n</i> = 37)	Reviewers (<i>n</i> = 13)	Critics (<i>n</i> = 98)	
Informal				$X^2 = 21.044, 8 \text{ df}$ $p = 0.0070$
Discussions	4.54	4.61	3.86	
Formal				$X^2 = 9.32, 8 \text{ df}$ $p = 0.3160$
Journals	3.67	3.92	3.83	
Conferences	3.40	3.69	3.29	$X^2 = 6.18, 8 \text{ df}$ $p = 0.6268$

TABLE 4. Formal and informal channel use by central and peripheral individuals.

	Formal Only	Informal Only	Both
Central individuals	—	28.3% (15)	71.7% (38)
Peripheral individuals	15.8% (15)	46.3% (44)	37.9% (36)

channels only). Patterns of use by the noncentral individuals were very different. Only slightly more than one-third—37.9%—used both types of boundary-spanning channels; nearly one-half (46.3%) used only the informal channels. Almost 16% used only the formal channels.

Thus, individuals who occupied central roles in the informal boundary-spanning network did not rely on personal discussion as the sole means of communication. Although their usage of informal channels was heavy, they were more likely than peripheral individuals to use formal media as well.

The central and peripheral respondents did not differ, however, in their perceptions of the importance of the various media for their work and thinking. They were asked to rate the importance of personal discussions, journal reading, and attendance at conferences and book fairs on scales from 5 (highly important) to 1 (not at all important). The mean ratings are displayed in Table 5; the ratings by both groups were highest for personal discussion, followed by reading journals, and finally attendance at conferences and book fairs. The central individuals consistently rated the three media higher than the peripherals, but the order of importance was the same. The differences between the two groups on this question were not statistically significant.

Boundary-Spanning Media Compared to Group Specific Media

The three professions differed in their preferences for boundary spanning as opposed to group-specific channels. Table 6 displays these differences.

The boundary-spanning media played the strongest role for the reviewers. Of their five most-used media, the first four are those which form the boundary-spanning structure. The four are also in the top five of the

editor's most-used media, but are used by a smaller percentage of the editors than of the reviewers, and even the highest is 15% lower than the nonboundary-spanning trade journal *Publishers Weekly*. The critics are the most isolated in their communication patterns; only two of the boundary-spanning media fell within their top five, the social circle in the highest position and *The Horn Book* were in the lowest.

Discussion

Communication is a fundamental process in the growth of knowledge. Communication within a discipline or profession establishes its identity, shapes the conduct of its work, and both identifies and disseminates its intellectual content. Communication across recognized boundaries imports ideas and information with the potential to modify the shared intellectual base.

The growth of knowledge is shaped in part by the specific processes and media employed. The research literature on boundary-spanning communication has most often focused on direct, personal contacts and overlooked the possible role of formal communication channels. This study of the members of three professions confirmed that boundary-spanning communication is indeed important to their work, and found that a variety of media were important components of the communication structure.

The structure was used by more than 80% of the respondents, indicating its importance in the work-related communication of the three professions. Both measures of importance—numbers of individuals linked and ratings of perceived importance—showed informal communication to be more important than formal, but also clearly indicated that formal channels are important sources of boundary-spanning communication which should not be overlooked.

TABLE 5. Mean rated importance of communication channels for central and peripheral individuals (5 = high, 1 = low).

	Discussion	Journals	Conferences
Central individuals	4.3	4.0	3.6
Peripheral individuals	4.0	3.6	3.2
	$X^2 = 6.69, 3 \text{ df},$ $p = 0.0825$	$X^2 = 9.35, 4 \text{ df},$ $p = 0.0528$	$X^2 = 6.174, 4 \text{ df},$ $p = 0.1900$

TABLE 6. Usage of the top five boundary-spanning and group-specific channels by all respondents.

Editors ($n = 37$):	<i>Publishers Weekly</i>	80%
	Social circle	66%
	<i>The Horn Book Magazine</i>	55%
	<i>School Library Journal</i>	55%
	Conferences of ALA	39%
Reviewers ($n = 13$):	Social circle	100%
	Conferences of ALA	92%
	<i>The Horn Book Magazine</i>	85%
	<i>School Library Journal</i>	85%
	<i>Bul. Ctr. for Ch. Books</i>	54%
Critics ($n = 98$):	Social circle	75%
	<i>Children's Literature</i>	44%
	<i>Children's Literature Association Quarterly</i>	42%
	Conferences of ChLA	36%
	<i>The Horn Book Magazine</i>	32%

Boundary-spanning communication, among the three professions, had the following features:

Importance as measured by number of individuals linked was greater for informal channels than formal, but each provided links between more than 50% of the cosmopolitans, and each linked individuals not reached by the other. The social circle linked 88.5% of the cosmopolitans, while the three formal channels linked 60.1%. The social circle was the exclusive boundary-spanning communication channel for 39.9%, the formal channels for 10.1%. When the professions are considered in the aggregate, informal channels hold greater importance on the measure of quantity, but the importance differs on this measure when the groups are considered separately.

Half of the individuals who used any boundary-spanning medium used informal and formal channels both; one-half used one or the other exclusively. While both types of channels are important as the sole boundary-spanning medium for many individuals, they are also apparently important when used together.

Perceived importance was somewhat higher for informal channels than for formal channels, and, of the two formal channels, journal reading was the more important. Personal discussion was rated in one of the two highest categories by 76.4% of the cosmopolitans, journal reading by 63.7%, and conference attendance by 48.6%.

The professions differed in which type of channel was most used, but were similar in their ratings of importance. The reviewers were the most comprehensive and balanced in their usage of the boundary-spanning

structure; 100% of the reviewers used channels of both types. Two-thirds of the editors also used both, but only one-third of the critics did so. The editors and critics were very different when those using only one type of boundary-spanning channel were considered; almost one-fourth of the editors used formal channels exclusively, but slightly less than 10% of the critics did so. The differences between groups were even more pronounced when informal channels were considered; more than half of the critics used informal channels only, but less than 10% of the editors did so. All three groups rated informal discussions the most highly, followed by journals, and then conferences.

Individuals central to the informal structure were also more likely to use the formal channels. It is likely that their central positions permit them to wick information from the formal channels into the system.

Boundary-spanning media were to be found among the five most frequently cited media for all groups, but the number ranged from four out of five to two out of five. For the editors, four of the five most frequently cited media spanned professional boundaries, but the one group-specific medium was the most heavily cited of the five. For the reviewers also, four of the five most frequently cited media were boundary spanning, but for them the group specific medium was the least cited of the five. For the critics, only two of the five most cited media were boundary spanning; one was the most cited, and the other the least cited of the five.

Each of these professional groups fills a separate cultural niche. Editors make literature for children publicly available; their job requires them to see literature

as a product as well as a work of art. Reviewers fill an intermediary role between producer and consumer; they provide evaluative information for other cultural intermediaries, librarians, and booksellers, who, in turn, make the books available to a third group of intermediaries (parents and teachers) and also to the ultimate consumers, children. Scholars stand outside the direct producer–consumer relationship. Their influence is primarily upon other scholars and, in many cases, university students. These students may become part of the producer–consumer relationship as parents, teachers, or librarians, or they may become part of the scholarly community and contribute to the growth of knowledge in this subdiscipline of the humanities.

The intellectual work of all three professions involves the analysis of cultural products. Though their work is focused upon specific artifacts, the surrounding cultural milieu of values and standards is potentially a part of each individual's intellectual environment. Many previous studies have linked boundary-spanning communication to vitality. The data of this study show that, in addition to the professions' internal communications and their members' individual knowledge, they share a boundary-spanning structure which links them to each other. The number of individuals included in the structure and their perceptions indicate that both the informal and the formal links to the other professions are highly important to their intellectual work.

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